

6

Molecular polarons

6.1 General features

A molecular polaron is defined as a polaron whose self-trapped electronic charge carrier is primarily confined within a single molecule. The molecule may be a constituent of a molecular solid, a solid comprised of molecules, e.g. pentacene. The molecule might also be an additive as occurs with molecularly doped polymers. Often, as in biological materials, the embedding medium contains displaceable ions and polar molecules.

The concept of a molecular polaron emerges naturally when a carrier's slow intermolecular motion fosters its self-trapping on a molecule. In particular, the difficulty of intermolecular motion is associated with modest intermolecular electronic overlap and correspondingly narrow electronic energy bands. By contrast, electronic transfer among at least several of a molecule's atoms may be rapid.

One can envision three types of molecular polaron. As schematically depicted in Fig. 6.1a, these three kinds of molecular polaron are distinguished by the spatial extent of their electronic carrier. The largest carrier extends over all of the equivalent structural units of the molecule. These structural units can be individual atoms, bonds or multi-atomic chemical units such as rings of atoms. A smaller carrier is limited to a non-minimal subset of the equivalent structural units. The smallest type of molecular polaron has its carrier collapsed to a single structural unit.

The three types of molecular polaron are analogous to the three solutions to the nonlinear eigenvalue equation Eq. (4.19) which governs an electronic carrier's self-trapping in a deformable continuum. These three solutions are those for a free carrier, a large polaron and a small polaron.

The scaling procedure of Section 4.3 provides an efficient means of establishing the three kinds of molecular polaron. In particular, this method is used to find the minima of the adiabatic potential for an electronic carrier occupying a deformable molecule embedded within a medium containing displaceable ions and/or polar molecules (Emin, 2000a). The electronic kinetic energy term drives the electronic

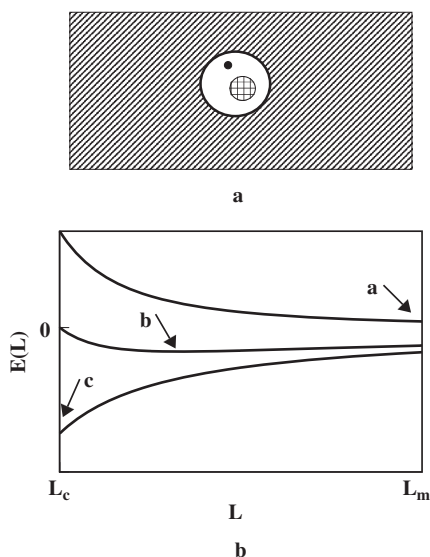


Fig. 6.1 Panel *a* is a schematic depiction of a molecule (white circle) embedded within a medium (diagonal-hatched area). Three types of molecular polaron are possible. (1) A molecular polaron's carrier encompasses all of the molecule's structures that feasibly can be occupied. (2) The carrier can occupy a non-minimal subset of these structures (square-hatched region within the molecule). (3) The self-trapped electronic carrier can collapse to a single structural unit (black dot). Panel *b* plots a molecular polaron's energy function $E(L)$ against its size parameter L for three values of the functional's physical parameters. The value L_m corresponds to the carrier's maximal expansion. The minimum value L_c indicates a carrier's collapse to its minimal size. The minimum value of $E(L)$ on each of the three curves is indicated with an arrow. Curve *a* depicts a maximally expanded carrier. Curve *b* corresponds to a carrier with intermediate expansion. Curve *c* indicates a collapsed carrier.

carrier to expand over the entire molecule. The electron–phonon interactions foster a carrier's localization. Here the short-range electron–phonon interaction describes the dependence of the energy of the carrier on displacements of the molecule's atoms that it contacts. The long-range Fröhlich electron–phonon interaction depicts the dependence of the carrier's potential energy on displacements of ions and polar molecules of the surrounding medium. The energy functional expressing the minimum of the adiabatic potential as a function of L , the scaling parameter characterizing the self-trapped carrier's spatial extent, is

$$E(L) = \frac{T_e}{L^2} - \frac{V_s}{L^d} - \frac{V_L}{L}, \quad (6.1)$$

where d is the dimensionality of the carrier's expansion.

The three types of minima of $E(L)$ with respect to L shown in Fig. 6.1b correspond to three kinds of molecular polaron. As illustrated by curve *a*, when the electronic kinetic energy is large enough the minimum of $E(L)$ occurs at L_m , the maximum physically acceptable value of L . In this instance the carrier extends over the entire molecule. This solution is like the free-carrier solution found in Chapter 4 except that the carrier's spatial extent is limited by the boundary of the molecule. As depicted in curve *b*, with a sufficiently strong long-range electron–phonon interaction the minimum of $E(L)$ falls at a value of L lying between that corresponding to the electronic carrier collapsing to a single site L_c and L_m . Then the electronic carrier will occupy a subset of the molecule's equivalent sites. This solution is analogous to that for a large polaron. Curve *c* depicts the situation in which the molecule's carrier has a very small electronic kinetic energy. The minimum of $E(L)$ is then at L_c . The electronic carrier then collapses to a single structural subunit of the molecule. This solution is analogous to that for a small polaron.

As with polaron formation generally, the formation of a molecular polaron is associated with carrier-induced shifts of atoms' equilibrium positions and their vibration frequencies. As described in Section 4.2, carrier-induced alterations of atomic vibration frequencies occur as the electronic carrier adjusts to atoms' vibratory movements. As noted below Eq. (4.18) this effect increases strongly with the localized electron's spatial extent. Therefore carrier-induced shifts of vibration frequencies tend to be especially large for molecular polarons. Electronic carriers associated with intermediate-sized molecular polarons may be most polarizable since they are neither constrained by the carrier's collapse nor its molecule's spatial extent.

Carrier-induced shifts of atoms' vibration frequencies alter atoms' vibration entropy. Carrier-induced shifts of atoms' vibration entropy affect the rates with which carriers' execute phonon-assisted hops and the entropy transported with these hops. These features of polaron hopping provide an explanation of the empirically observed Meyer–Neldel compensation effect and the enhancement of carriers' Seebeck coefficients. These effects are discussed in Chapter 11 and Chapter 12, respectively.

The activation energies of polaron jump rates generally rise with jump distance. By itself, this effect fosters carriers making short hops. Applying an electric field promotes carriers making especially long jumps in concert with it. These two effects compete with one another. For a polaron formed through its carrier's long-range Coulomb-based electron–phonon interaction with ions and polar molecules surrounding it, the trade-off can produce a contribution to a polaron's hopping activation energy which falls in proportion to the square root of the electric-field strength. This effect is termed Frenkel–Poole behavior. It is also addressed in Chapter 11.

Holstein introduced the molecular-crystal model (MCM) as a simple pedagogic construct to address some aspects of polaron formation and small-polaron hopping

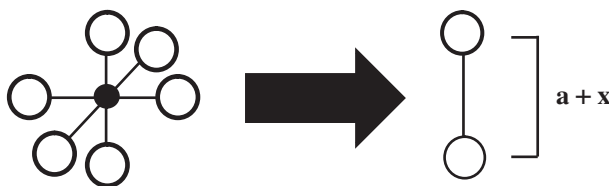


Fig. 6.2 A transition-metal cation (e.g. Mn in MnO) depicted as a black dot surrounded by six displaceable oxygen ligands (open circles) is represented in the molecular-crystal model as a diatomic molecule whose solitary deformation coordinate x is the deviation of its interatomic separation from its carrier-free equilibrium value a .

(Holstein, 1959a; 1959b). However, this model does not represent a real molecular solid. Its pseudo-molecules do not permit carrier-induced softening. It also ignores the long-range electron–phonon interaction characteristic of an electronic carrier in a polar medium. Indeed, the term polaron was coined to emphasize the importance of these interactions to self-trapping phenomena in polar and ionic media. As noted above, these features can generate the Meyer–Neldel and Frenkel–Poole effects.

The molecular-crystal model (MCM) envisions a regular array of pseudo-molecules that each has but one scalar deformation coordinate (Holstein, 1959a). In particular, the deformation coordinate of a pseudo-molecule at position $\mathbf{g}$ is denoted as $x_{\mathbf{g}}$. The energy of a carrier on this molecule is taken to be $\varepsilon_{\mathbf{g}} = -Fx_{\mathbf{g}}$. Thus, adding a carrier to the molecule shifts the equilibrium value of its deformation coordinate without affecting its vibration frequency. This model was developed to provide an idealized depiction of self-trapping in a simple inorganic solid. For example, as illustrated in Fig. 6.2, an oversimplified model for the small-polaron hole in MnO presumes that the energy of a severely localized carrier on a transition-metal cation only depends on the positions of its nearest-neighbor oxygen ligands. The small polaron forms as the ligands undergo symmetry-breaking (Jahn–Teller) displacements about the Mn^{3+} hole thereby lifting its electronic degeneracy. The MCM represents this carrier-induced displacement of oxygen ions' equilibrium positions as that induced by a carrier occupying a non-degenerate electronic state on its pseudo-molecule's single scalar deformation coordinate.

6.2 Examples

Polarons formed on asymmetric molecules are widely studied. In particular, because of electro-photographic applications considerable effort has been devoted to studying polaron motion between asymmetric molecules embedded among polymers. For example, holes are thought to hop between *p*-diethylaminobenzaldehyde diphenylhydrazone (DEH) molecules dispersed among polycarbonate and polystyrene polymers

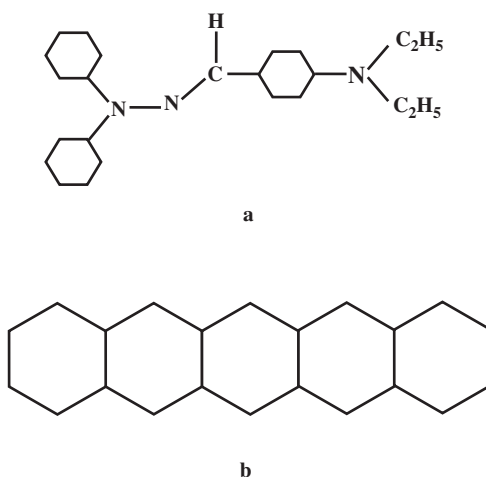


Fig. 6.3 Panel *a* depicts a DEH molecule. Panel *b* illustrates a pentacene molecule.

(Schein and Borsenberger, 1993). As illustrated in Fig. 6.3a, a DEH molecule is composed of chemically distinct subunits. A self-trapped carrier is then presumably confined to some subunit (e.g. a planar ring) of this multi-component molecule. Then maximal electronic overlap between corresponding structures of neighboring molecules requires their alignment with one another. As a result, intermolecular transport can be tortuous among a disordered distribution of such dopant molecules embedded within a medium.

Polarons also may form on large molecules comprising a symmetric array of nearly equivalent structures. For example, the linear pentacene molecule, illustrated in Fig. 6.3b, is based on the merger of five benzene molecules. In these situations charge may be distributed nearly uniformly over the chemically equivalent portions of the molecule. As described in Section 4.2, intramolecular movement of a weakly localized charge carrier in response to atomic motion is then prone to significantly shift the stiffness and frequencies of the associated atomic vibrations. As will be discussed in Chapters 11 and 12, carrier-induced changes in atomic frequencies can make significant contributions to a charge carrier's mobility and Seebeck coefficient, respectively. Indeed, the large Seebeck coefficients observed in pentacene transistors have been attributed to carrier-induced changes of the stiffness of associated atomic vibrations (von Mühlénen *et al.*, 2007).